

2019 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Impulse and Momentum — Kicking a football

A footballer claims that a football can be kicked a greater distance if it is moving towards the player rather than being stationary. Use physics to comment.

Worked solution

The distance travelled depends on the ball's speed after the kick.

Impulse $J = F \Delta t = \Delta p = m(v_{\text{after}} - v_{\text{before}})$.

- If the ball is stationary, $\Delta p = m v_{\text{after}}$.
- If the ball is moving towards the player, then v_{before} is negative. So $\Delta p = m(v_{\text{after}} - (-v_{\text{before}})) = m(v_{\text{after}} + |v_{\text{before}}|)$. This is a larger change in momentum, so the impulse is greater.

For the same force and contact time, the final speed of the ball is greater. Therefore, the kicked ball will travel further if it was initially moving towards the player.

Final answer

Yes — because when the ball approaches the player, the change in momentum is greater, so the impulse imparted is larger and the final

speed (and distance) is greater.

Revision tips

- Impulse $J = F\Delta t = \Delta p$.
- Change in momentum depends on both final and initial velocities.
- If initial velocity is towards player, Δp is larger for same kick.
- Greater $\Delta p \Rightarrow$ greater $v_{\text{after}} \Rightarrow$ longer distance travelled.
- Always consider velocity direction (signs) when calculating Δp .